

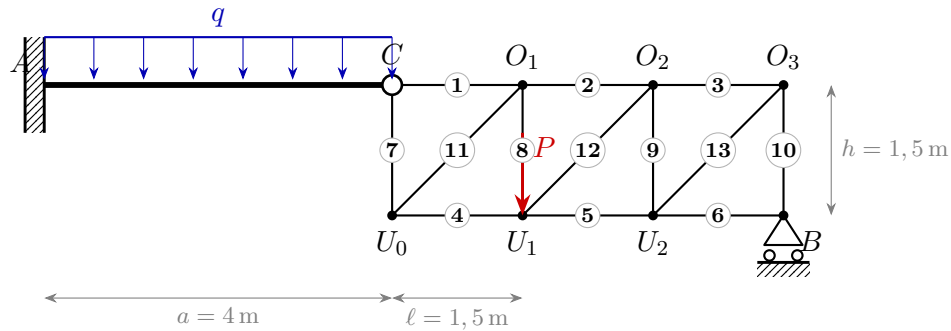
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VU Mechanik – LAWI 100515 (PI)

Exam **Group A**

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Model solution



1. Static determinacy

Counting criterion $n = r - (3 + v)$ with r support reactions and v hinge conditions:

$$n = r - (3 + v) = 4 - (3 + 1) = 0 \quad \Rightarrow \quad \text{the system is statically determinate.}$$

2. Support reactions and hinge force

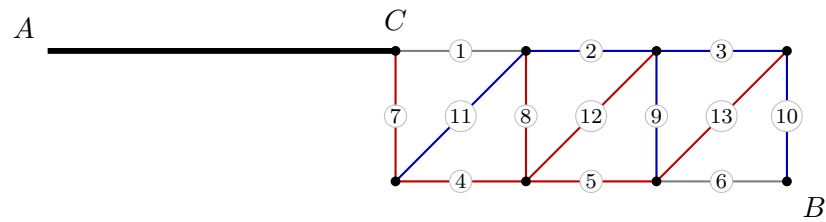
$$A_x = 0 \text{ kN}, \quad A_y = 30.67 \text{ kN}, \quad M_A = 74.67 \text{ kNm}, \quad B_y = 3.33 \text{ kN}$$

$$C_x = 0 \text{ kN}, \quad C_y = -6.67 \text{ kN}$$

3. Truss member forces

Member	Member force S [kN]	Type
1 (C–O1)	0	zero-force
2 (O1–O2)	-6.67	compression
3 (O2–O3)	-3.33	compression
4 (U0–U1)	6.67	tension
5 (U1–U2)	3.33	tension
6 (U2–U3)	0	zero-force
7 (C–U0)	6.67	tension
8 (O1–U1)	6.67	tension
9 (O2–U2)	-3.33	compression
10 (O3–U3)	-3.33	compression
11 (U0–O1)	-9.43	compression
12 (U1–O2)	4.71	tension
13 (U2–O3)	4.71	tension

red: tension (+) blue: compression (–)



4. Section-force diagrams of the bending beam A–C

